FULLY FAITHFUL ADJOINTS DETECTED BY THE UNIT AND COUNIT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3669-3686, following the statement at 3656-3667.

Editorial note. The dual second part (line 3685, phrase: part (2) is dual to part (1)) is retained verbatim as a mirror-level omission; the executed part (1) covers its dual.

Allowed standard prerequisites (editorial).

- Adjunction bijection. definition-adjoint, cited at statement line 3658. Supplies the bijection Mor(u(X'), u(X)) = Mor(X', v(u(X))) used at proof lines 3672-3674.
- Left adjoint composed with fully faithful is fully faithful. lemma-left-adjoint-composed-fully-faithful, cited at proof lines 3678-3679. Supplies that v composed with u fully faithful forces u fully faithful in the second implication of part (1).

Source reference. Original definition 0037: sources/stacks-categories/pinned-original-source.tex, source lines 3561–3561. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FWV: sources/stacks-categories/pinned-original-source.tex, source lines 3625–3625. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let u be a left adjoint to v as in Definition 0037. Then*

(1) *u is fully faithful $\Leftrightarrow$ $id \cong v \circ u$ $\Leftrightarrow$ $\eta : id \to v \circ u$ is an isomorphism,*
(2) *v is fully faithful $\Leftrightarrow$ $u \circ v \cong id$ $\Leftrightarrow$ $\epsilon : u \circ v \to id$ is an isomorphism.*

Proof. Proof of (1). Assume u is fully faithful. We will show $\eta_X : X \to v(u(X))$ is an isomorphism. Let $X' \to v(u(X))$ be any morphism. By adjointness this corresponds to a morphism $u(X') \to u(X)$. By fully faithfulness of u this corresponds to a unique morphism $X' \to X$. Thus we see that post-composing by η_X defines a bijection $\mathrm{Mor}(X', X) \to \mathrm{Mor}(X', v(u(X)))$. Hence η_X is an isomorphism. If there exists an isomorphism $\mathrm{id} \cong v \circ u$ of functors, then $v \circ u$ is fully faithful. By Lemma 0FWV we see that u is fully faithful. By the above this implies η is an isomorphism.

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Thus all 3 conditions are equivalent (and these conditions are also equivalent to $v \circ u$ being fully faithful).

Part (2) is dual to part (1). $\square$